

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281021

Kind Code

A1

Publication Date

September 11, 2025

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DISHWASHER

Abstract

Provided is a dishwasher. The dishwasher includes an outer door; an inner door having a first through hole; and an air discharge device disposed between the outer door and the inner door, the air discharge device including an air duct and a fan, the air duct being in communication with a washing chamber of the dishwasher through the first through hole, the air duct having a water return opening upstream of the fan, and the water return opening being configured to allow external air to enter the air duct from the water return opening during an operation of the fan and to discharge condensate water into the washing chamber.

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Appl. No.: 19/214901

Filed: May 21, 2025

Foreign Application Priority Data

CN 202211468315.X

Nov. 22, 2022

CN 202223111273.2

Nov. 22, 2022

Related U.S. Application Data

parent WO continuation PCT/CN2023/097599 20230531 PENDING child US 19214901

Publication Classification

Int. Cl.: A47L15/48 (20060101); A47L15/42 (20060101)

U.S. Cl.:

CPC A47L15/488 (20130101); A47L15/4223 (20130101); A47L15/4246 (20130101); A47L15/4257 (20130101); A47L15/483 (20130101); A47L15/486 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/097599 filed on May 31, 2023, which is based on and claims priorities to Chinese Patent Application Nos. 202211468315.X and 202223111273.2 both filed on Nov. 22, 2022, the entire contents of each of which are incorporated herein by reference. No new matter has been introduced.

FIELD

[0002] The present disclosure relates to the field of cleaning device technologies, and more particularly, to a dishwasher.

BACKGROUND

[0003] In a drying phase, a dishwasher needs to use an air discharge device to extract air from a washing chamber, thereby drying the dishes. However, in an early stage of the drying phase, air in the washing chamber is at a high temperature and high humidity. Directly extracting the air from the washing chamber with the air discharge device results in the expelled air being at a high temperature, which poses a predetermined risk, and leads to generation of a significant amount of condensate water at an air discharge opening or in an external environment.

SUMMARY

[0004] The present disclosure aims to solve technical problems in the related art to some extent. To this end, the present disclosure provides a dishwasher, which is capable of effectively reducing a temperature of the expelled air and a probability of condensate water forming at an air discharge opening and outside the dishwasher.

[0005] To achieve the above objective, the present disclosure provides a dishwasher. The dishwasher includes: an outer door; an inner door connected to the outer door and having a first through hole; and an air discharge device disposed between the outer door and the inner door, the air discharge device including an air duct and a fan, the air duct being in communication with a washing chamber of the dishwasher through the first through hole, the fan being adapted to draw air from the washing chamber into the air duct and discharge the air out of the air duct, the air duct having a water return opening upstream of the fan, and the water return opening being configured to allow external air to enter the air duct from the water return opening during an operation of the fan and to discharge condensate water into the washing chamber.

[0006] In some embodiments of the present disclosure, the dishwasher includes a tub. The tub includes a bottom plate enclosing the washing chamber. The water return opening is located above the bottom plate and spaced apart from the bottom plate.

[0007] In some embodiments of the present disclosure, the water return opening is located directly above the bottom plate.

[0008] In some embodiments of the present disclosure, the condensate water drips from the water return opening onto the bottom plate.

[0009] In some embodiments of the present disclosure, the water return opening is open

downwardly.

[0010] In some embodiments of the present disclosure, the water return opening is in communication with a space between the inner door and the outer door.

[0011] In some embodiments of the present disclosure, the water return opening is located below an inlet of the fan.

[0012] In some embodiments of the present disclosure, an open direction of the water return opening spatially intersects with an air intake direction of the fan.

[0013] In some embodiments of the present disclosure, the water return opening is arranged adjacent to the fan.

[0014] In some embodiments of the present disclosure, the air duct includes a condensation segment extending in an up-down direction, the water return opening being located below the condensation segment.

[0015] In some embodiments of the present disclosure, an open direction of the water return opening is the same as an extension direction of the condensation segment.

[0016] In some embodiments of the present disclosure, the air duct includes an air intake segment, a condensation segment, and an air discharge segment that are sequentially in communication with each other, the air intake segment being constructed as an inverted U-shape and in communication with the washing chamber, the fan being disposed in the air discharge segment, and the water return opening being disposed in one of the air discharge segment and the condensation segment.

[0017] In some embodiments of the present disclosure, the air duct includes an air intake tube forming the air intake segment, a condensate tube forming the condensate segment, and an air discharge tube forming the air discharge segment, and the fan and the water return opening being located in the air discharge segment.

[0018] In some embodiments of the present disclosure, the fan is located below $\frac{1}{2}$ or $\frac{1}{3}$ of a height of the inner door, and the air discharge segment has an air discharge opening located below the inner door and the outer door.

[0019] In technical solutions of the present disclosure, with the air discharge device between the outer door and the inner door, the air in the washing chamber is effectively extracted and discharged, and thus drying of the dishes is realized. The air discharge device has the water return opening. The water return opening is configured to discharge the condensate water, and when the fan is in operation, the external air can enter the air duct through the water return opening and mix with high-temperature and high-humidity air extracted into the washing chamber in the air duct. In this way, an air discharge temperature of the air discharge device and a risk can be reduced, and moisture in the air when the air is mixed can be further removed. The formed condensate water is discharged back to the washing chamber through the water return opening, so as to reduce a probability of generation of the condensate water in an external environment. By disposing the water return opening, both recirculation of condensate water and mixing of air to lower the air discharge temperature are realized.

[0020] Additional aspects and advantages of the present disclosure will be provided at least in part in the following description, or will become apparent at least in part from the following description, or can be learned from practicing of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and/or additional aspects and advantages of the present disclosure will become more apparent and more understandable from the following description of embodiments taken in conjunction with the accompanying drawings.

[0022] FIG. 1 is a schematic view of mounting of an air discharge device according to some

embodiments of the present disclosure.

[0023] FIG. 2 is a schematic view of mounting of an air discharge device according to some embodiments of the present disclosure.

[0024] FIG. 3 is a schematic view of an air discharge device according to some embodiments of the present disclosure.

[0025] FIG. 4 is an enlarged view of a region indicated by a dashed line in FIG. 3.

[0026] FIG. 5 is a schematic view of an air intake segment/an air intake tube according to some embodiments of the present disclosure.

[0027] FIG. 6 is a schematic view of a condensation segment/a condensation tube according to some embodiments of the present disclosure.

REFERENCE NUMERALS OF THE ACCOMPANYING DRAWINGS

[0028] door body **1000**; inner door **1100**; first through hole **1110**; outer door **1200**; [0029] air discharge device **2000**; [0030] air duct **2100**; air intake segment/air intake tube **2110**; air intake opening **2111**; condensation segment/condensation tube **2120**; air discharge segment/air discharge tube **2130**; air discharge opening **2131**; water return opening **2132**; [0031] fan **2200**; [0032] tub **3000**; washing chamber **3100**; bottom plate **3110**; rear side plate **3120**; top plate **3130**.

DETAILED DESCRIPTION OF EMBODIMENTS

[0033] Technical solutions according to embodiments of the present disclosure will be described clearly and completely below in combination with accompanying drawings of the embodiments of the present disclosure. Obviously, the embodiments described below are only a part of the embodiments of the present disclosure, rather than all embodiments of the present disclosure.

Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative labor fall within the protection scope of the present disclosure.

[0034] It should be noted that all directional indications (such as up, down, left, right, front, rear, etc.) in the embodiments of the present disclosure are only used to explain relative positions between various components, movements of various components, or the like under a predetermined posture (as illustrated in the figures). When the predetermined posture changes, the directional indications also change accordingly.

[0035] In the present disclosure, unless otherwise clearly specified and limited, terms, such as “connect” “fix” or the like, should be understood in a broad sense. For example, it may be a fixed connection or a detachable connection or connection as one piece; mechanical connection or electrical connection; direct connection or indirect connection through an intermediate; internal communication of two components or the interaction relationship between two components, unless otherwise clearly limited. For those of ordinary skill in the art, the specific meaning of the above-mentioned terms in the present disclosure can be understood according to specific circumstances.

[0036] In addition, descriptions, such as “first” “second” or the like, in the present disclosure are only used for descriptive purposes, and cannot be understood as indicating or implying relative importance or implicitly indicating the number of indicated technical features. Therefore, the features associated with “first” and “second” may explicitly or implicitly include at least one of the features. In addition, combinations can be performed on the technical solutions according to various embodiments of the present disclosure, but these combinations must be based on the fact that they can be realized by those skilled in the art. When a combination of the technical solutions is contradictory or unattainable, the combination of the technical solutions neither exists nor falls within the protection scope of the appended claims of the present disclosure.

[0037] A dishwasher is a device that can automatically clean dishes. Generally speaking, the dishwasher includes a base, a tub, and a door body. The tub is mounted on the base. The tub has a washing chamber. The door body is rotatably connected to the tub and/or the base for closing or opening the washing chamber. The washing chamber is provided with a dish rack that can be pulled out or pushed in. When the door body opens, the user can pull out the dish rack, load the dishes onto the dish rack, and then push the dish rack back into the washing chamber after loading. The

user then closes the door body to seal the washing chamber, and the washing chamber can begin a washing process.

[0038] A spray arm is disposed in the washing chamber. The spray arm is in communication with an outlet of a washing pump of the base through a water path, and an inlet of the washing pump is in communication with a water cup. The water cup is located at a bottom of the tub and in communication with the washing chamber. Water in the washing chamber can flow to the water cup. When the dishwasher begins to operate, a water inlet valve of the dishwasher opens, allowing water to flow into the washing chamber. Once a water level in the washing chamber reaches a predetermined level, the water inlet valve closes, cutting off continued input of water. The dishwasher controls the washing pump to operate. The washing pump sucks water in the water cup and delivers the water to the spray arm. The spray arm has a nozzle. The water flow is ejected from the nozzle of the spray arm under an action of a pressure. The spray arm is urged to rotate under an action of a reverse driving force of the water flow ejected from the nozzle of the spray arm, in such a manner that the water flow is ejected onto the dishes to clean the dishes. The water sprayed by the spray arm flows back to the washing chamber and then flows to the water cup. The water cup is provided with a filter. The water flow continues to circulate under an action of the washing pump after being filtered by the filter until a current washing sequence is finished.

[0039] To improve a cleaning effect of the dishes, generally speaking, it is necessary to heat the water during the washing sequence. For example, a heating assembly is disposed in the washing pump. The washing pump may synchronously heat the water while sucking the water in the washing chamber. That is, the water flow sprayed by the spray arm has a higher temperature, improving the cleaning effect on the dishes.

[0040] After a washing phase, the dishes need to be dried. Since the washing chamber of the tub is in a high-temperature and high-humidity environment after the washing stage is finished, in the related art, a dishwasher is provided with the air discharge device. The air discharge device directly extracts the air in the washing chamber and discharges the air to an external environment.

[0041] Since the air in the washing chamber in this case is at a high temperature and high humidity, the air discharged by the air discharge device has a high temperature, which poses a burn risk. In addition, when the high-temperature and high-humidity air is discharged to the external environment and encounters cooler temperatures, condensate water can form. For example, the condensate water can form at an air discharge outlet of the air discharge device. The condensate water can run down an exterior surface of the dishwasher to the floor. Or, when the expelled air encounters an obstacle, the condensate water can form on the obstacle, such as on the floor, which not only diminishes user experience, but also poses a safety risk. Therefore, the present disclosure provides a dishwasher, which is capable of reducing a probability of the condensate water forming outside the dishwasher, and reducing the air discharge temperature, and thus improving the user experience.

[0042] As illustrated in FIG. 1 to FIG. 4, in some embodiments of the present disclosure, the dishwasher includes a door body **1000** as a structure configured to close a tub **3000**. The door body includes an outer door **1200** and an inner door **1100**. The inner door **1100** and the outer door **1200** are connected to each other through some connection methods. For example, the outer door **1200** and the inner door **1100** are superimposed together, and a periphery of the inner door **1100** and a periphery of the outer door **1200** are connected to each other through screwing. Generally speaking, the inner door **1100** and the outer door **1200** are sheet metal parts, which are suitable for use in larger dishwashers and have high structural strength. However, the inner door **1100** and the outer door **1200** are not limited to the sheet metal parts, and may be made of other materials. For example, when the dishwasher is a small desktop machine, the inner door **1100** and the outer door **1200** may be made of plastic.

[0043] The inner door **1100** has a first through hole **1110**. For example, the first through hole **1110** is a circular through hole. An air discharge device **2000** can be in communication with a washing

chamber **3100** through the first through hole **1110**, extracting air from the washing chamber **3100**. It should be understood that the air discharge device **2000** in communication with the washing chamber **3100** through the first through hole **1110** can be that an air intake opening **2111** of the air discharge device **2000** passes through the first through hole **1110** to be in communication with the washing chamber **3100**, or that the air intake opening **2111** of the air discharge device **2000** may be connected to the inner door **1100** to allow the air intake opening **2111** of the air discharge device **2000** to be in communication with the first through hole **1110**, and the first through hole **1110** is in communication with the washing chamber **3100**, realizing communication between the air intake opening **2111** of the air discharge device **2000** and the washing chamber **3100**.

[0044] The air discharge device **2000** is a device configured to extract and discharge the air in the washing chamber **3100**. The air discharge device **2000** is mounted between the inner door **1100** and the outer door **1200**. That is, after the inner door **1100** is connected to the outer door **1200**, a predetermined space between the inner door **1100** and the outer door **1200** is formed to place the air discharge device **2000**. The air discharge device **2000** may be fixed only to the inner door **1100**, may also be fixed only to the outer door **1200**, or may also be fixed to both the inner door **1100** and the outer door **1200**.

[0045] The air discharge device **2000** includes a fan **2200** which provides power for the air discharge device **2000** to extract and discharge the air in the washing chamber **3100**, and an air duct **2100** which forms a path through which air flows. In an exemplary embodiment of the present disclosure, the air duct **2100** needs to be disposed to be in communication with the washing chamber **3100**. As described above, the air duct **2100** may pass through the first through hole **1110** to achieve communication with the washing chamber **3100**. For example, the air duct **2100** has the air intake opening **2111** in communication with the washing chamber **3100**. The air duct **2100** has an air discharge opening **2131**, through which the air duct **2100** is in communication with the external environment. The fan **2200** is disposed in the air duct **2100**. When the fan **2200** is in operation, a negative pressure may be formed in the air duct **2100**. Therefore, the air in the washing chamber **3100** can enter the air duct **2100** through the air intake opening **2111**, then flow through the fan **2200** and continue to be discharged to the external environment through the air discharge opening **2131** of the air duct **2100**. It should be understood that the fan **2200** may have various positions in the air duct **2100**. For example, the fan **2200** is arranged close to the air intake opening **2111** and away from the air discharge opening **2131**, that is, close to the first through hole **1110**. Or, the fan **2200** may be arranged close to the air discharge opening **2131** and away from the air intake opening **2111**, that is, away from the first through hole **1110**. It is also possible that the fan **2200** is arranged between the air intake opening **2111** and the air discharge opening **2131** of the air duct **2100**, as long as the air in the washing chamber **3100** can be extracted into the air duct **2100** and discharged from the air duct **2100**.

[0046] The air duct **2100** has a water return opening **2132** configured to discharge the condensate water back into the washing chamber **3100**. Since the high-temperature and high-humidity air in the washing chamber **3100** enters the air duct **2100**, which is equivalent to the high-temperature and high-humidity air being cold, a predetermined amount of condensate water is formed in the air duct **2100**. When the condensate water in the air duct **2100** is not treated in time, an efficiency of air extraction and discharge can be affected, and even the condensate water can flow out of the air discharge opening **2131**. Therefore, with the water return opening **2132**, when the condensate water is formed in the air duct **2100**, the condensate water flows into the washing chamber **3100** through the water return opening **2132**, and thus the condensate water flowing back into the washing chamber **3100** can be further discharged by a drain pump of the dishwasher. It should be understood that the tub **3000** of the dishwasher forms the washing chamber **3100**. The water cup is disposed below the tub **3000** of the dishwasher, and in communication with the washing chamber **3100**. That is, the condensate water flowing out of the water return opening **2132** may eventually flow to the water cup and then be drained away through drainage. Or, the condensate water flowing

out of the water return opening **2132** can directly flow to the water cup by some middle conduits. Since the water cup is in communication with the washing chamber **3100**, this situation is also regarded as the condensate water being discharged to the washing chamber **3100**.

[0047] In addition to discharging the condensate water into the washing chamber **3100**, the water return opening **2132** is also designed to allow external air to freely enter the air duct **2100**.

[0048] In particular, the water return opening **2132** is disposed upstream of the fan **2200**. It should be understood that, relative to the fan **2200**, the fan **2200** has an inlet and an outlet. Operation of the fan **2200** makes air enter from the inlet and be discharged from the outlet. A path through which the air flows to the inlet can be regarded as an upstream, and a path through which the air flows out from the outlet can be regarded as a downstream. Since the water return opening **2132** is disposed upstream of the fan **2200**, when the fan **2200** is in operation, the air in the air duct **2100** has a fast flow rate and a smaller pressure. Therefore, the external air (which has a lower temperature and humidity than the air in the washing chamber **3100**) can enter the air duct **2100** from the water return opening **2132** to mix with the high-temperature and high-humidity air from the washing chamber **3100** in the air duct **2100**, realizing mixing of air to reduce the air discharge temperature, preventing the air discharge temperature from being too high and causing harm to the user, especially young children.

[0049] In addition, relative to the high-temperature and high-humidity air in the washing chamber **3100**, the external air has a relatively lower temperature and lower humidity. In this way, when the two air streams are mixed in the air duct **2100**, a predetermined amount of condensate water may occur, which can also be discharged from the air duct **2100** through the water return opening **2132** to avoid flowing out at the air discharge opening **2131**. Since the high-temperature and high-humidity air from the washing chamber **3100** can reduce moisture in the air after being mixed with the external air, the air discharged through the air discharge opening **2131** is less likely to form the condensate water in the external environment, reducing a probability of forming the condensate water in the external environment, improving the user experience.

[0050] The fan **2200** extracts the air in the washing chamber **3100** and discharges the air to the external environment through the air discharge opening **2131**. The air discharge opening **2131** may be mounted at various positions, such as drilling a hole on a front surface of the outer door **1200** for the air discharge opening **2131** to discharge the air, or drilling a hole on a side surface of the outer door **1200** for the air discharge opening **2131** to discharge the air. However, these discharging air methods involve drilling a hole in the user's line of sight, which is likely to compromise overall integrity of the door body **1000**. For this reason, in another exemplary embodiment of the present disclosure, the air discharge opening **2131** of the air discharge device **2000** is disposed below the door body **1000**, for example, between the door body **1000** and a toe kick board. That is, the dishwasher has the door body **1000** and the toe kick board located below the door body **1000** at a front side of the dishwasher. The toe kick board is disposed to avoid being too hollowed out under the door body **1000**, and the dishwasher has a more complete front side. Since the door body **1000** needs to be set as rotatable, there is a predetermined gap between the door body **1000** and the toe kick board. The air discharge opening **2131** may be in the gap between the door body **1000** and the toe kick board and opened towards a front of the dishwasher (a side close to the user is the front side when the user uses the dishwasher, that is, an orientation where the door body **1000** is located is the front side, and the door body **1000** rotates from top to bottom for opening the washing chamber **3100**). In this way, the air discharge opening **2131** is hidden in the user's line of sight without affecting air discharge.

[0051] As can be understood, the fan **2200** is a device capable of providing power for extracting and discharging the air. For example, the fan **2200** includes a motor and an impeller drivingly connected to the motor. The impeller is disposed in the air duct **2100** to drive the air to flow through the air duct **2100**. In another exemplary embodiment of the present disclosure, the impeller adopts a structure of a centrifugal impeller to allow the air duct **2100** to be made flatter, reducing a

space between the outer door **1200** and the inner door **1100** occupied by the air discharge device **2000**.

[0052] As can be seen from the above, by providing the air discharge device **2000** between the outer door **1200** and the inner door **1100**, the air in the washing chamber **3100** is effectively extracted and discharged, and thus the drying of the dishes is realized. The air discharge device **2000** has the water return opening **2132**. The water return opening **2132** is configured to discharge the condensate water, and when the fan **2200** is in operation, the external air can enter the air duct **2100** through the water return opening **2132** and mix with the high-temperature and high-humidity air extracted into the washing chamber **3100** in the air duct **2100**. In this way, the air discharge temperature of the air discharge device **2000** and a risk can be reduced, and moisture in the air when the air is mixed can be further removed. The formed condensate water is discharged back to the washing chamber **3100** through the water return opening **2132**, so as to reduce the probability of generation of the condensate water in the external environment. By disposing the water return opening **2132**, both recirculation of condensate water and mixing of air to lower the air discharge temperature are realized.

[0053] As illustrated in FIG. **3** and FIG. **4**, in some embodiments of the present disclosure, to facilitate outflow of the condensate water from the water return opening **2132**, the water return opening **2132** is open downwardly in this embodiment. Where the “open downwardly” can be directly below or obliquely below, and the “open” means forming an opening to be in communication with the air duct **2100**. In this way, when the condensate water is formed in the air duct **2100**, the condensate water can flow out from the water return opening **2132** under the action of gravity, without a need for additional guide structures or extraction devices for extracting. This design prevents accumulation of the condensate water and makes an overall structure of the air discharge device **2000** simpler and more reliable, thereby reducing cost of the product.

[0054] For example, when the air from the washing chamber **3100** enters the air duct **2100**, the high-temperature and high-humidity air encounters a wall of the air duct **2100** and is cooled to form the condensate water on the wall of the air duct **2100**, and the condensate water flows along the wall of the air duct **2100** to the water return opening **2132** under the action of gravity and is discharged through the water return opening **2132**. However, the air entering from the water return opening **2132** and the high-temperature and high-humidity air from the washing chamber **3100** collide and create turbulence in the air duct **2100**, which causes the condensate water to form on corresponding walls of the air duct **2100** and flow out through the water return opening **2132** under the action of gravity.

[0055] It should be understood that when the condensate water is discharged from the water return opening **2132**, the air in the air duct **2100** is still in a flowing state in this case. Therefore, the external air can enter the air duct **2100** from the water return opening **2132** under the action of a pressure. That is, discharging of the condensate water and entry of the external air are performed simultaneously. The discharging of the condensate water does not affect the entry of the external air into the air duct **2100** through the water return opening **2132**, and the entry of the external air into the air duct **2100** through the water return opening **2132** does not affect the discharging of the condensate water, in such a manner that the extraction efficiency of the air in the washing chamber **3100** can be improved, and thus a drying efficiency of the dishes can be improved.

[0056] In some embodiments of the present disclosure, as illustrated in FIG. **1** and FIG. **2**, the dishwasher includes the tub **3000** forming the washing chamber **3100**. For example, the dishwasher has a top plate **3130**, a rear side plate **3120**, a left side plate, a right side plate, and a bottom plate **3110**. The left side plate is disposed at a side of the rear side plate **3120**. The right side plate is disposed at another side of the rear side plate **3120**. The top plate **3130** is disposed at a top of the rear side plate **3120**, the left side plate, and the right side plate. The bottom plate **3110** is disposed at a bottom of the rear side plate **3120**, the left side plate, and the right side plate. In this way, a washing chamber **3100** that is open towards the front is formed. The door body **1000** is rotatably

disposed in front of the tub **3000** for opening or closing the washing chamber **3100**. In this way, the water return opening **2132** is disposed above the bottom plate **3110** and spaced apart from the bottom plate **3110**, in such a manner that the condensate water flowing out of the water return opening **2132** can be received by the bottom plate **3110** from top to bottom to flow into the washing chamber **3100**, which is simple and convenient and effectively realizes backflow of the condensate water.

[0057] In an exemplary embodiment of the present disclosure, the water return opening **2132** is disposed above the bottom plate **3110**. The “above” here may be directly above or obliquely above, as long as it is ensured that the condensate water flows to the bottom plate **3110** under the action of gravity and is received by the bottom plate **3110**. For example, when the water return opening **2132** is located directly above the bottom plate **3110**, that is, in a top-down projection of the dishwasher, a projection of the water return opening **2132** is located in a projection of the bottom plate **3110**, a front end of the bottom plate **3110** may extend forwardly to extend between the inner door **1100** and the outer door **1200**. In this way, the condensate water from the water return opening **2132** can flow back into the bottom plate **3110** in a shorter and faster path and be received by the bottom plate **3110**, improving a discharge efficiency of the condensate water and avoiding generation of the condensate water in the door body **1000** due to the external air exchanging heat with the condensate water during the condensate water flowing back to the washing chamber **3100**.

[0058] When the water return opening **2132** is not located directly above the bottom plate **3110** but is located obliquely above the bottom plate **3110**, backflow of the condensate water to the washing chamber **3100** can also be realized, which is slightly inconvenient as compared to a situation where the water return opening **2132** is located directly above the bottom plate **3110**. For example, in the top-down projection of the dishwasher, the projection of the water return opening **2132** is not located in the projection of the bottom plate **3110**, but in front of the projection of the bottom plate **3110**, in such a manner that the water return opening **2132** needs to guide the condensate water onto the bottom plate **3110** through some guide structures to be received by the bottom plate **3110**. This more complicated guide structure may increase design difficulty of the air discharge device **2000**, and may occupy too much space of the outer door **1200** and the inner door **1100**, and even make a thickness of the door body **1000** increase.

[0059] In another exemplary embodiment of the present disclosure, as illustrated in FIG. 1 and FIG. 2, in some embodiments, the water return opening **2132** is designed to be spaced apart from the bottom plate **3110** without any direct contact, which further simplifies the structure, and avoids increasing complexity of the air discharge device **2000** by mounting of intermediate pieces, enabling the condensate water to return to the washing chamber **3100** at a fastest speed, and thus avoiding condensation of air in the door body **1000** during the backflow of the condensate water.

[0060] For example, when the water return opening **2132** is located directly above the bottom plate **3110**, the water return opening **2132** may be designed to be close to the top of the bottom plate **3110** and in non-contact with the bottom plate **3110**. In this way, the condensate water flowing out of the water return opening **2132** is separated from the water return opening **2132** under the action of gravity and drips onto the bottom plate **3110**. The condensate water is only subjected to resistance of air but not by other resistance in a process of dripping onto the bottom plate **3110**, improving the backflow efficiency of the condensate water and avoiding condensation of air in the door body **1000**. In addition, after a drying stage of the dishwasher is completed, there is no residual condensation water in a condensation water dripping path from the water return opening **2132** onto the bottom plate **3110**, and there is no possibility that condensation water flows to other parts of the door body **1000** after the door body **1000** of the dishwasher is rotated and opened.

[0061] For another example, when the water return opening **2132** is located obliquely above the bottom plate **3110**, a guide groove needs to be designed between the water return opening **2132** and the bottom plate **3110**. The condensate water dripping from the water return opening **2132** drips to the guide groove and is guided to the bottom plate **3110** through the guide groove. Compared to a

scenario where the condensate water drips directly onto the bottom plate **3110**, this method introduces an additional guide groove, which creates resistance to flow of the condensate water, reducing the backflow efficiency of the condensate water. In addition, this method increases complexity of the outer door **1200** and the inner door **1100** to some extent.

[0062] As described above, to reduce a temperature and humidity of the high-temperature and high-humidity air from the washing chamber **3100** in the air duct **2100**, the entry of the external air is realized by disposing the water return opening **2132**. It should be understood that the external air has a lower temperature and lower humidity than the high-temperature and high-humidity air of the washing chamber **3100**. To facilitate the entry of the external air into the air duct **2100** through the water return opening **2132**, as illustrated in FIG. 1 and FIG. 2, in some embodiments, the water return opening **2132** is in communication with the space between the outer door **1200** and the inner door **1100**. That is, the water return opening **2132** is disposed between the inner door **1100** and the outer door **1200** and in direction communication with the space between the inner door **1100** and the outer door **1200**. Further, generally speaking, the inner door **1100** and the outer door **1200** of the dishwasher are not sealed together. That is, the space enclosed by the inner door **1100** and the outer door **1200** is in communication with an external space outside the dishwasher. Therefore, when the air enters the air duct **2100** through the water return opening **2132**, air from the external space outside the dishwasher may also enter the space between the inner door **1100** and the outer door **1200**.

[0063] The water return opening **2132** is in communication with the space between the outer door **1200** and the inner door **1100**, in such a manner that the entire air discharge device **2000** can be hidden between the inner door **1100** and the outer door **1200**. An existing non-sealed connection structure between the outer door **1200** and the inner door **1100** can be fully utilized to realize entry of cold air. An opening on the outer door **1200** can be avoided to make the cold air enter the air duct **2100**, resulting in a simpler structure and aesthetic appearance of the door body **1000**.

[0064] In another exemplary embodiment of the present disclosure, as illustrated in FIG. 3 and FIG. 4, in some embodiments, the water return opening **2132** is disposed below the inlet of the fan **2200**, which is more conducive to the discharging of the condensate water and prevents the condensate water from flowing out by the fan **2200**.

[0065] For example, the condensate water is formed on the wall of the air duct **2100**. The condensate water may flow towards the water return opening **2132** under the action of gravity, and then flow out through the water return opening **2132**. Since the inlet of the fan **2200** has a higher position than the water return opening **2132**, the condensate water is more likely to cross the inlet of the fan **2200** and flow through the water return opening **2132** under the action of gravity, and less likely to form convergence at the inlet of the fan **2200**, avoiding entering the fan **2200**.

[0066] As illustrated in FIG. 3 and FIG. 4, in some embodiments, an open direction of the water return opening **2232** spatially intersects with an air intake direction of the fan **2200**, to prevent the condensate water from entering the fan **2200** to the greatest extent.

[0067] For example, when the condensate water is formed in the air duct **2100** and flows to the water return opening **2132** in a direction of gravity, a flow direction of the air in the air duct **2100** is roughly the same as a flow direction of the condensate water under an action of the fan **2200**. That is, flow of the air in the air duct **2100** causes the condensate water to flow faster. Since the air intake direction of the fan **2200** intersects with the open direction of the water return opening **2132**, when the condensate water flows to a position opposite to the fan **2200**, it is equivalent to a sudden change in the intake air of the fan **2200**. In this way, the condensate water can continue to flow to the water return opening **2132** under an action of inertia, reducing a likelihood of the condensate water being carried into the fan **2200** by an airflow. This design prevents the condensate water from being discharged through the air discharge opening **2131**. For example, as illustrated in FIG. 3 and FIG. 4, the open direction of the water return opening **2132** is from top to bottom, and the air intake direction of the fan **2200** is perpendicular to a plane of paper/screen.

[0068] It should be understood that the fan **2200** includes the impeller, for example, a centrifugal impeller. The air intake direction of the fan **2200** is an extension direction of a rotational axis of the centrifugal impeller.

[0069] In another exemplary embodiment of the present disclosure, a main purpose of the fan **2200** is to extract the air in the washing chamber **3100**. To effectively introduce the external air into the air duct **2100** through the water return opening **2132**, the water return opening **2132** is arranged adjacent to the fan **2200**, in such a manner that when the fan **2200** operates, a path from the water return opening **2132** to the inlet of the fan **2200** is shortened, enabling the external air to be more effectively introduced into the air duct **2100**. Here, adjacent arrangement can be understood as being close to each other, for example, 3 cm, 5 cm, 8 cm, etc. apart.

[0070] As mentioned above, the external air fed into the air duct **2100** through the water return opening **2132** has the lower temperature and the lower humidity, which can neutralize the air having a higher temperature and higher humidity from the washing chamber **3100** in the air duct **2100**. However, the air from the washing chamber **3100** has a large amount under the action of the fan **2200**. If the amount of the external air entering from the water return opening **2132** is to be increased, a size of the water return opening **2132** needs to be increased. In this case, the extract effect on the washing chamber **3100** may be reduced. Therefore, before the air from the washing chamber **3100** meets the external air entering through the water return opening **2132**, the air needs to be condensed. This pre-reduction of the temperature and humidity of the air from the washing chamber **3100** ensures that, when the air mixes with the external air from the water return opening **2132**, the temperature and the humidity of the air can be minimized as much as possible. As illustrated in FIG. 3, in some embodiments, the air duct **2100** includes a condensation segment **2120**. The so-called condensation segment **2120** can make the air from the washing chamber **3100** be condensed to a certain extent, reducing moisture and a temperature in the air. That is, the condensation segment **2120** needs to be provided with a condensation structure. For example, a structure of the condensation segment **2120** is set as a concave-convex shape to increase a heat exchange area, and thus the air in the air duct **2100** is effectively condensed.

[0071] The condensation segment **2120** extends in an up-down direction, in such a manner that the condensate water generated in the condensation segment **2120** flows downwardly under the action of gravity. When the water return opening **2132** is disposed below the condensation segment **2120**, the condensate water can more easily flow to the water return opening **2132** and be discharged. In another exemplary embodiment of the present disclosure, an open direction of the water return opening **2132** is the same as an extension direction of the condensation segment **2120**. In this way, the condensate water can be directly discharged through the water return opening **2132**, preventing the condensate water from forming a circuitous path and accelerating the discharge efficiency of the condensate water. For example, as illustrated in FIG. 3 and FIG. 6, the condensation segment **2120** has a long strip structure extending in the up-down direction. The water return opening **2132** is disposed below the condensation segment **2120** and opens downwardly, in such a manner that the condensate water can flow out of the water return opening **2132** from top to bottom. In particular, when the condensate water generated in the condensation segment **2120** is separated from the wall of the condensation segment **2120**, the condensate water can be directly discharged through the water return opening **2132** under the action of the gravity and the airflow, effectively accelerating a drainage efficiency of the condensate water.

[0072] As illustrated in FIG. 3, FIG. 5 and FIG. 6, in some embodiments, the air duct **2100** includes an air intake segment **2110**, a condensation segment **2120**, and an air discharge segment **2130** that are sequentially in communication with each other. The air intake segment **2110** has the air intake opening **2111**. The air discharge segment **2130** has the air discharge opening **2131**. The air intake segment **2110** is in communication with the washing chamber **3100** through the air intake opening **2111**. The fan **2200** is disposed in the air discharge segment **2130**. When the fan **2200** is in operation, the air from the washing chamber **3100** enters the air intake segment **2110**, then flows

through the condensation segment **2120** and the air discharge segment **2130**, and is ultimately discharged. The air intake segment **2110** is constructed as an inverted U-shape. In this way, when the dishwasher is in the washing state, even if the water flow sprayed by the spray arm of the dishwasher enters the air intake segment **2110** through the air intake opening **2111**, the water entering through the air intake opening **2111** will flow back into the washing chamber **3100** instead of entering the condensation segment **2120**, because the air intake segment **2110** is an inverted U-shape, and a part of the air intake segment **2110** having the air intake opening **2111** extends from bottom to top.

[0073] The water return opening **2132** may be disposed in the air discharge segment **2130** or the condensation segment **2120**. Since the main function of the condensation segment **2120** is condensation, the water return opening **2132** disposed in the air discharge segment **2130** can avoid adverse effects on air condensation.

[0074] In an exemplary embodiment of the present disclosure, the air duct **2100** includes an air intake tube **2110** forming the air intake segment **2110**, a condensate tube **2120** forming the condensate segment **2120**, and an air discharge tube **2130** forming the air discharge segment **2130**. The air intake tube **2110**, the condensate tube **2120**, and the air discharge tube **2130** are manufactured separately and then connected together. The air intake tube **2110** and the condensate tube **2120**, and the condensate tube **2120** and the air discharge tube **2130** can be connected to each other by a variety of connection methods, such as insertion, snap, welding, or screwing, which makes manufacturing of the entire air duct **2100** more convenient. For the air intake tube **2110**, the condensate tube **2120**, and the air discharge tube **2130**, each can be formed by connecting two pieces, for example, by welding the two pieces. In this way, a hollow structure for the air flow can be created.

[0075] Through condensation of the condensation segment **2120** and introducing the external air through the water return opening **2132**, the moisture of the air discharged from the air duct **2100** is greatly reduced, and a probability of forming condensate water in the air discharge opening **2131** and in the external environment is greatly reduced. However, the final discharged air still contains moisture and cannot achieve complete drying. To further reduce formation of the condensate water, as illustrated in FIG. 2, in some embodiments, the fan **2200** is located below $\frac{1}{2}$ or $\frac{1}{3}$ of a height of the inner door **1100**. That is, a distance from the fan **2200** to a bottom edge of the inner door **1100** (excluding hinges on two sides of the inner door **1100**) is smaller than or equal to $\frac{1}{2}$ or $\frac{1}{3}$ of the height of the inner door **1100**, resulting in a shorter path from the fan **2200** to the air discharge opening **2131**. The shorter path reduces a probability of air condensing in this path, allowing the air to be discharged more quickly into the external environment to further avoid the probability of formation of the condensate water at the air discharge opening **2131**. In addition, the air discharged into the external environment, having significantly reduced moisture, is less likely to form the condensate water when the air mixes thoroughly with the external air.

[0076] Although some embodiments of the present disclosure are described above, the scope of the present disclosure is not limited to the embodiments. Under the concept of the present disclosure, any equivalent structure transformation made using the contents of the specification and the accompanying drawings of the present disclosure, or any direct or indirect application of the contents of the specification and the accompanying drawings of the present disclosure in other related fields, shall equally fall within the scope of the present disclosure.

Claims

1. A dishwasher, comprising: an outer door; an inner door connected to the outer door and having a first through hole; and an air discharge device disposed between the outer door and the inner door, the air discharge device comprising an air duct and a fan, the air duct being in communication with a washing chamber of the dishwasher through the first through hole, the fan being adapted to draw

- air from the washing chamber into the air duct and discharge the air out of the air duct, the air duct having a water return opening upstream of the fan, and the water return opening being configured to allow external air to enter the air duct from the water return opening during an operation of the fan and to discharge condensate water into the washing chamber.
2. The dishwasher according to claim 1, further comprising a tub, the tub comprising a bottom plate enclosing the washing chamber, and the water return opening being located above the bottom plate and spaced apart from the bottom plate.
 3. The dishwasher according to claim 2, wherein the water return opening is located directly above the bottom plate.
 4. The dishwasher according to claim 3, wherein the condensate water drips from the water return opening onto the bottom plate.
 5. The dishwasher according to claim 1, wherein the water return opening is open downwardly.
 6. The dishwasher according to claim 1, wherein the water return opening is in communication with a space between the inner door and the outer door.
 7. The dishwasher according to claim 1, wherein: the water return opening is located below an inlet of the fan; and/or an open direction of the water return opening spatially intersects with an air intake direction of the fan.
 8. The dishwasher according to claim 1, wherein the water return opening is arranged adjacent to the fan.
 9. The dishwasher according to claim 1, wherein the air duct includes a condensation segment extending in an up-down direction, the water return opening being located below the condensation segment.
 10. The dishwasher according to claim 9, wherein an open direction of the water return opening is the same as an extension direction of the condensation segment.
 11. The dishwasher according to claim 1, wherein the air duct comprises an air intake segment, a condensation segment, and an air discharge segment that are sequentially in communication with each other, the air intake segment being constructed as an inverted U-shape and in communication with the washing chamber, the fan being disposed in the air discharge segment, and the water return opening being disposed in one of the air discharge segment and the condensation segment.
 12. The dishwasher according to claim 11, wherein the air duct comprises an air intake tube forming the air intake segment, a condensate tube forming the condensation segment, and an air discharge tube forming the air discharge segment, and the fan and the water return opening being located in the air discharge segment.
 13. The dishwasher according to claim 11, wherein: the fan is located below $\frac{1}{2}$ or $\frac{1}{3}$ of a height of the inner door; and the air discharge segment has an air discharge opening located below the inner door and the outer door.
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